

Advanced ML, DL, NLP Course

①

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Sec 3D Xg boost ML Algorithms

Lec 1 :- Xg Boost Classification
Algorithm

Lec 2 :- Xg Boost Regression

Dataset

Salary	Credit	Approval
<= 50K	B	0
<= 50K	G	1
<= 50K	G	1
<= 50K	B	0
> 50K	G	1
> 50K	G	0
> 50K	G	0
<= 50K	G	0

Steps

- (1) Construct a base model
- (2) Construct a Decision Tree with root.
- (3) calculate similarity matrix

$$= \frac{\sum (\text{Residual } P)}{\sum P_i (-P_i) + \lambda}$$

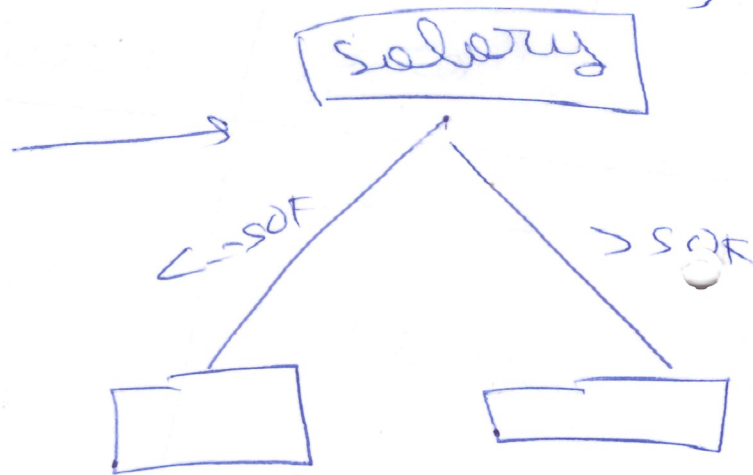
- (4) calculate Gain

$[-0.5, 0.5, 0.5, 0.5, -0.5]$

(1)



$P_i = 0.5$



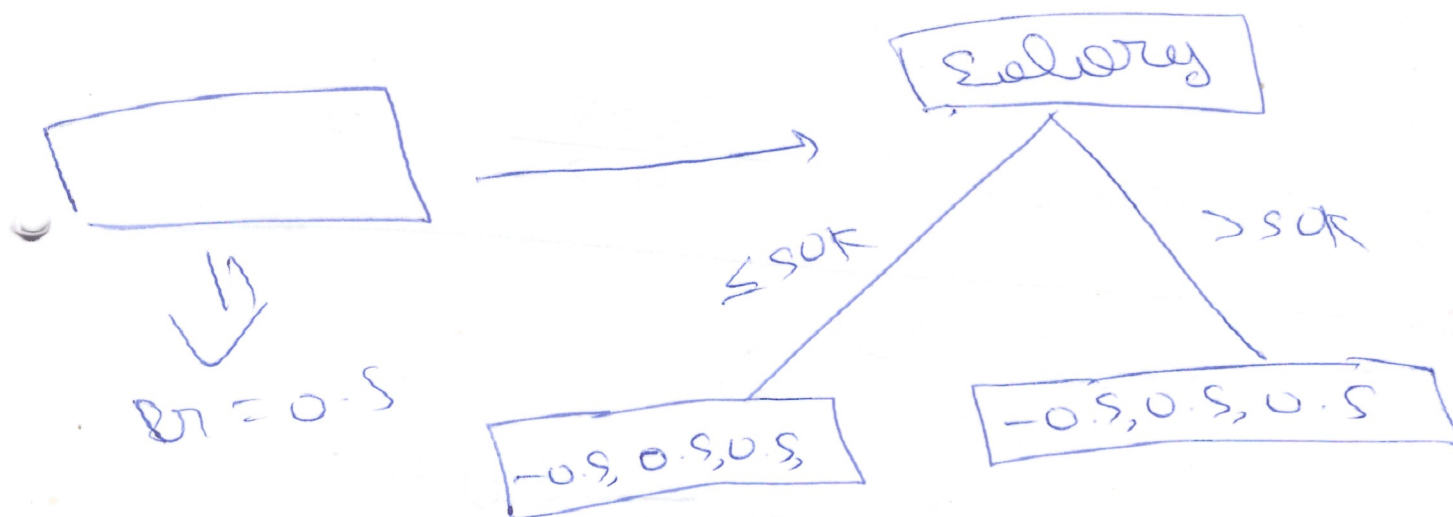
(3)

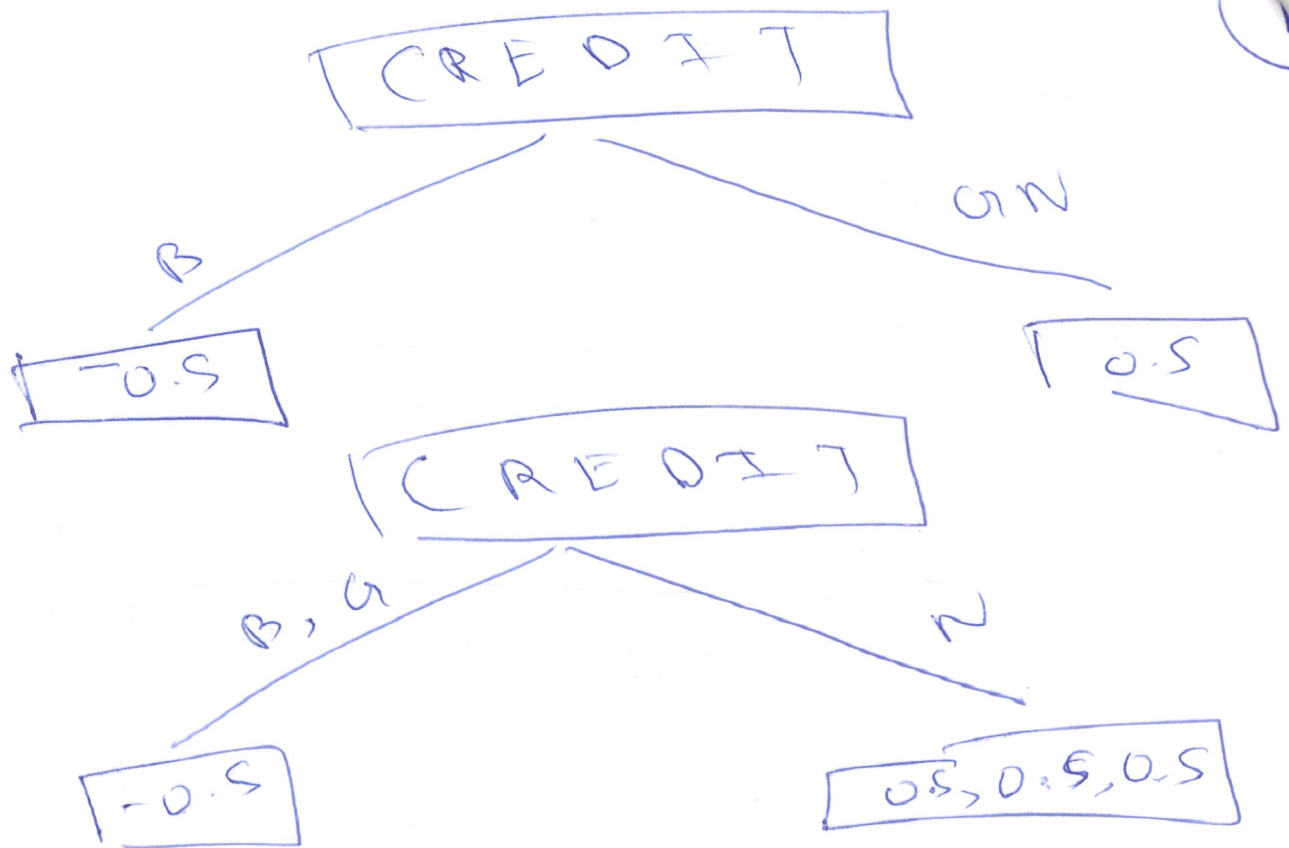
Similarity
weight (α_c)

$$= \frac{\sum (\text{residual})^2}{\sum R^2 (1 - R^2)}$$

$$= \frac{[(-0.5 + 0.5 + 0.5 - 0.5)^2]}{[0.5 \cdot (1 - 0.5) + 0.5 \cdot (1) + 0.5 \cdot (1) + 0.5 \cdot (1)]}$$

$$= 0$$





$$\text{Similarity (dc)} = \frac{0.25}{0.25} = 1$$

$$\text{Similarity} = \frac{0.25}{0.25} = 0.33$$

$\Rightarrow$ Final O/P

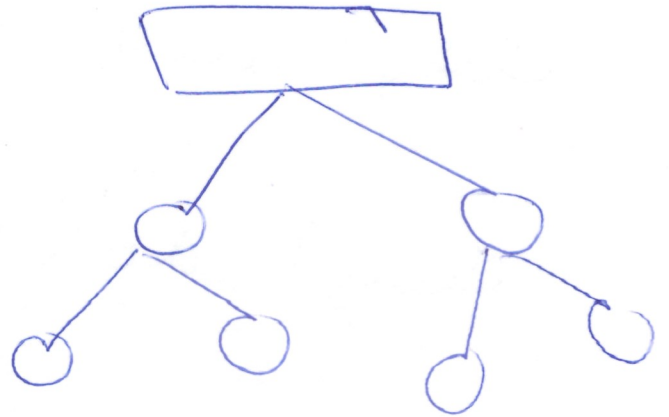
$$\text{New test ratio} = 0 +$$

Final O/P

5



$$O/P = 0$$



plus
data

$$= \sigma(0 + d(1))$$

↓
Sigmoid
Activation

↓
Similarly
there

$$d = 0.1$$

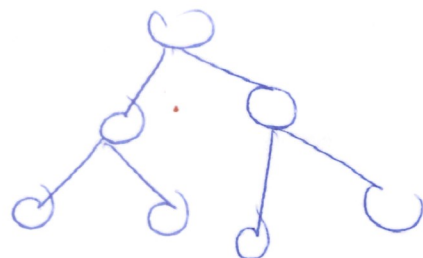
$$= \sigma(0 + (0.1)(1))$$

$$= \frac{1}{1 + e^{-0.1}} = 0.52$$

$$= \frac{1}{1 + e^{-0.088}} = 0.508$$



d



test data

(6)

$$\log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

$$\log(\text{odds}) = \log\left(\frac{0.5}{0.5}\right) = 0$$

$$\text{Output}(0|1) = \sigma(\text{Bias learned})$$

$$+ \alpha_1(p\tau_1) +$$

$$\alpha_2(p\tau_2) + \alpha_3(p\tau_3)$$

calculate similarity
weight

$$= \frac{\sum (\text{residual})^2}{\sum p_i(1-p_i) + \lambda_2}$$

⇒ Xy Boost regressor
ML algorithm

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Dataset

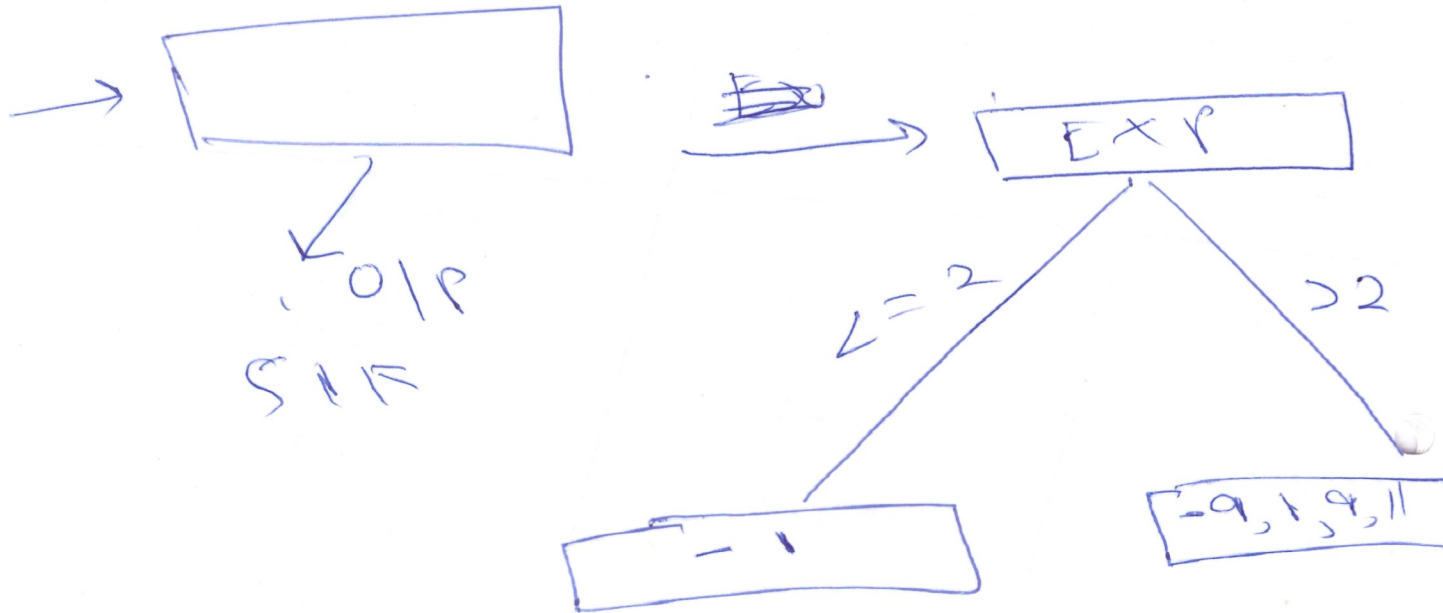
Edn	Crash	Salary
2	yes	40K
2.5	yes	42K
3	no	52K
4	no	60K
4.5	yes	62K

$$\text{Similarity weight} = \frac{\sum (\text{residual } R)}{\sum R(1-R)}$$

Steps

(8)

(1) Create a Base Model



$$\text{Stripping weight} = \frac{\sum (\text{Residual})^2}{\text{No. of Residuals}}$$

$$\text{sw (left child)} = \frac{121}{14}$$

↓ ↑ sw

sw (right child)

$$= \frac{(-9 + 1 + 9 + 1)^2}{4 - 1}$$

$$= \frac{144}{3}$$

$$= 28.8$$

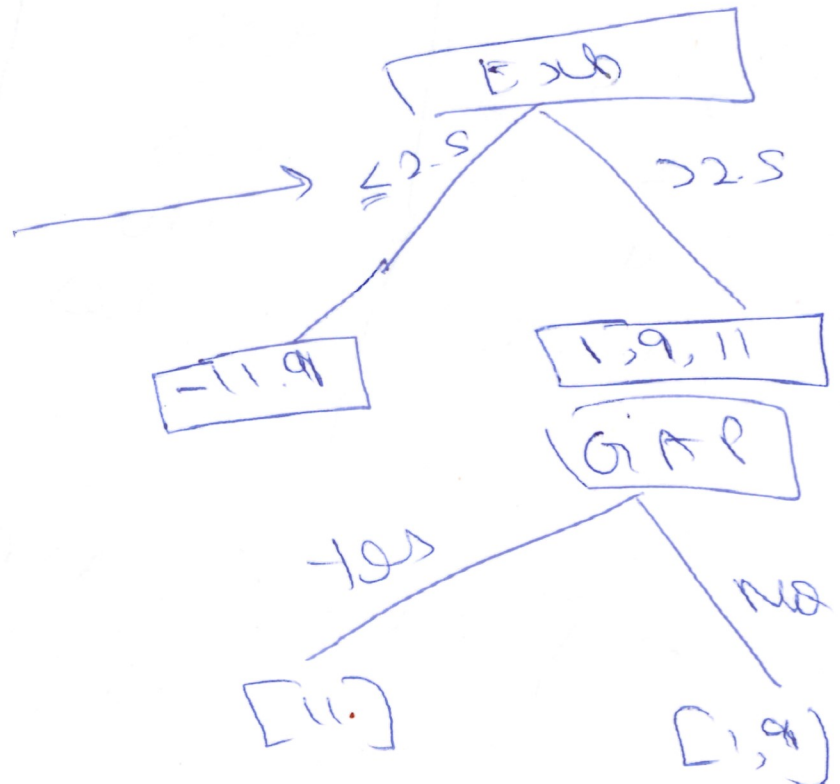
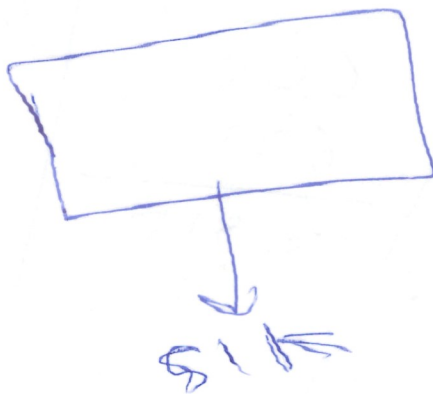
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③ Calculate Crain

$$\begin{aligned} \text{Crain} &= 65.5 \times 28.5 - 0.16 \\ &= 98.34 \end{aligned}$$



$$\text{Crain} = 14, 3, 42$$



XG Boost Classification

(10)

→ Best learner +

$$\alpha_1(DT_1) + \alpha_2(DT_2) + \dots + \alpha(DT_n)$$

XG Boost regressor = $S1K + 0.1(S)$

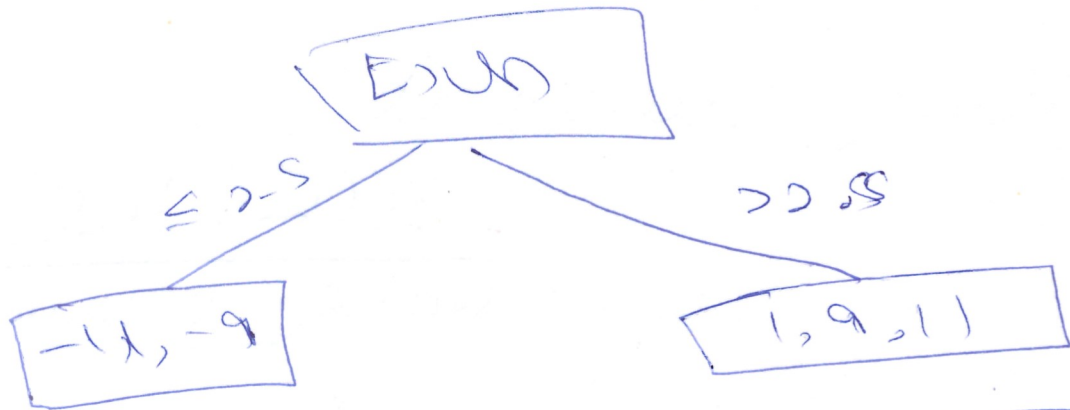
$$= \$1 + 0.5$$

$$= \$1.5$$

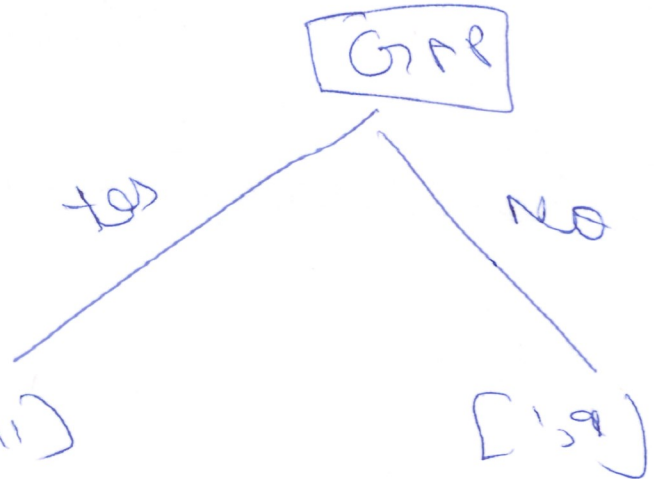
Ex	Crash	Salary	P_1	$\hat{y}$
2	yes	40K	-11	
2.5	yes	42K	-9	
3	yes	52K	1	
4	no	60K	9	
4.5	yes	62K	11	
		$\approx 51K$		

$$= \$1 + (0.1)(-10)$$

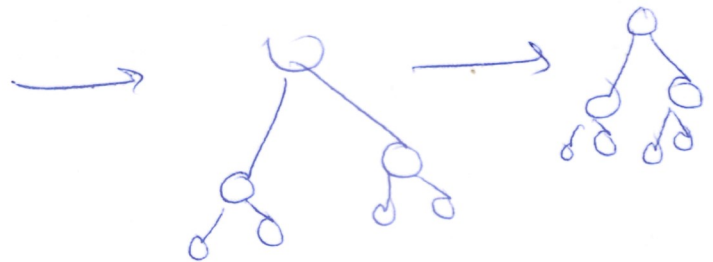
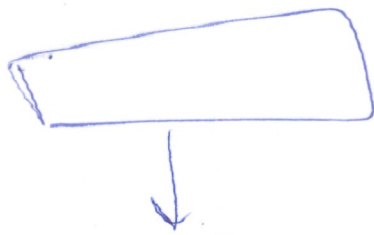
$$= \$1 - 0.1 = 0.9$$



$\swarrow$
 $O/P = [-11, -9]$



$$[5] + 0.7(11) = 5 + 7.7 = 12.7$$



11
 9
 7
 5
 3
 1

Similarly weight

$$= \frac{\sum (\text{Residual})^2}{\text{no. of residual} + 1}$$

of regression

$$\text{Similarly weight} = \frac{\sum (\text{residual})^2}{\sum R^2 (1 - R^2) + 1}$$